

Modern Geometry II course notes (Instructor Version)

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“It is not knowledge, but the act of learning, not possession but the act of getting there, which grants the greatest enjoyment.”

—Karl Friedrich Gauss, 1808

Acknowledgements

The materials used in these course notes have been drawn from a variety of sources. The notes are meant to be accompanied by the book *Experiencing Geometry In Euclidean, Spherical, and Hyperbolic Spaces* [Henderson & Tamiņa, 2005] and they incorporate many ideas from this book. Several of the problems from this book are reprinted in Appendix C. Section 4 explores the non-Euclid java application created by Joel Castellanos, Joe Dan Austin, and Ervan Darnell. When working on Problems 8.3 and 8.4, I encourage students to use Spherical Easel, a java application created by David Austin and William Dickinson. As discussed in Section 6, we usually also spend some time exploring the program KaleidoTile, which was created by Jeff Weeks. The activity in Section 7 is a slightly modified version of Project 64 from the booklet *101 Project Ideas for The Geometers Sketchpad* [Bennett, 1996]. The assignment in Section 9 discusses an excerpt from *Gödel, Escher, Bach: an Eternal Golden Braid* [Hofstadter, 1999] which is reprinted in Appendix D.

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Preface

“The royal road to knowledge,
it is easy to express:
to err, and err, and err again,
but less, and less, and less.”

This is the second course in a two course sequence that focuses on geometry and learning to develop and express your own mathematical ideas. Like the first course, it will emphasize ideas and imagination in addition to techniques and calculations. We will be further investigating Euclidean and Spherical Geometry, as well as other geometries, such as Taxicab Geometry and Hyperbolic Geometry. Like the first semester course, this course will focus on mathematical reasoning and communication skills in addition to mathematical facts.

These notes are meant to be accompanied by the textbook *Experiencing Geometry: Euclidean and Non-Euclidean with History* by David W. Henderson and Daina Taimina [Henderson & Taimina, 2005]. A lot of this book may already look familiar to you if you already took a course using the first part of these notes. Normally, the first semester of this course covers much of the material in chapters 1–4, 6, and 7. This course will look at significant parts of chapters 5, 8, and 10, and some other sections.

Introduction for the Instructor

This set of notes is an attempt to write down what happens in my second semester college geometry course, Math 342: Modern Geometry II, at the University of Northern Colorado. These materials are designed for students who have already been through a one semester modern geometry course at the college level. The materials work best with students who have taken a first semester class that used my materials, although quite a few students have successfully taken this class after taking some other first semester geometry class. The materials for my first semester class can be found in Issue No. 17 of this *Journal* [Miller, 2010]. I encourage anyone thinking about using these notes to also look at the course notes for that course. Much of the way that the class is run is similar to that class.

There are two major differences between the first and second semester classes as I have taught them. The first is that the second semester course is only required for pre-service secondary teachers, so that it is not generally taken by the pre-service elementary teachers that make up a large part of the students in the first semester course. The second is that for the second semester course, I ask the students to buy the book *Experiencing Geometry* [Henderson & Tamiņa, 2005] to use as a textbook for the course. Throughout these notes, whenever I refer to Problem x.x, I am referring to this book. The statements of the problems from *Experiencing Geometry* that are referred to in these notes are reprinted in Appendix C. However, the statements of these problems here are not intended to replace the fuller discussion of them that can be found in the original book.

In this instructor's version of these notes, I have included both the materials that I give to the students and also some teaching notes intended for course instructors. A separate student version of these notes contains only the student materials. Sections of these notes that appear in both versions have the same section number in both versions. In this instructor's version, many of the assignments are followed by teaching notes intended for course instructors. These notes are in blue, and do not appear in the student version. In addition, this instructor's version also includes this "Introduction for the Instructor," as well as several extra appendices of materials that instructors may find useful.

The following are some of the policies that I put in my syllabus. It is, of course, possible to use some or all of these assignments in a course with significantly different policies. See the course notes for the first semester class for further general details about how I run these classes.

Assignments: As in Modern Geometry I, there will be a variety of types of assignments in this course. Most of the assignments will be writing assignments. Some of these will be formal writing assignments, given letter grades, while others will be informal assignments. The informal assignments will go through a revision process and will be graded according to their completeness, as in Modern Geometry I. Some of them will be group assignments, and some will be individual assignments. There will also be a take-home midterm and a final project, which will be presented during our scheduled final exam time.

At the end of the semester, I will again ask you to turn in your portfolio containing all

the work that you have done for this class, including all drafts of all papers, so please save everything. On average, you should expect to spend at least nine hours per week outside of class on this course. If you are concerned about the time that you are spending on this class, come see me.

Due dates: All assignments will have assigned initial due dates. Informal assignments can be revised as many times as you like; however, they should be substantially done within sixteen days of the original due date. Progress made after this date will count half as much as progress made before this date, unless the problem is at least $3/4$ complete by then.

Course Materials: You will need a folder or binder in which to keep your written work, and some kind of disk on which to keep your work done on the computers. You will also want to have a sphere of some kind to look at when we study spherical geometry. Tennis balls work well, as do the plastic spheres often available at craft shops.

Group Work: We will often work in groups in this course. Whenever a group hands in a written assignment, they are required to put on the paper the names of those who participated fully, and only those names. Each person must sign the final copy. Your signature certifies that you participated equally in the project. It is dishonest to turn in work that is not solely and equitably the creation of the team members. You are not required to include on the report the name of someone who started but did not finish, or who did not contribute their share.

Outside Sources: A central aim of this course is to help you learn to develop your own ideas about mathematical questions. You therefore should NEVER consult any reference materials outside of the course text and materials in answering questions for this course. This includes materials found on the internet. The ideas that you present should be your own.

Further General Teaching Notes

Because this class is non-traditional, some further notes on how the class is run are included here. I encourage new teachers of these materials to consider these ideas, but to find a way to run the class that works for you and your teaching style.

The course is structured around the class assignments. Typically, for each assignment, I start by talking generally about the assignment and putting it into the context of other things we've been doing, and/or with a whole class discussion of some ideas leading into the assignment.

We generally then work on the assignment in groups. In my experience, groups of three work significantly better than groups of two or four. On the one hand, groups of three generate more ideas than groups of two; and on the other hand, it is easy for group members to be left out or to not fully participate in a group of four. I try to change the groups frequently, so that everyone in the class works together at one time or another. Student comments about the class indicate that while students are often hesitant at first about working in groups and about changing groups so often, once they see how productive it can be it ends up being one of their favorite aspects of the class. I have found that the easiest way to quickly form new random groups is by having the class count off (by the number of people in class divided by the number of people per group) and then assign each group number to a specific table or area of the room. The class works best when it meets in a room with tables, with enough tables that each group can have a table to itself.

While the groups are working, I walk from group to group and discuss with them what they are doing. If they are stuck, I ask them to tell me about what they have tried so far,

and try to ask questions that will lead them in a productive direction. In any case, I try to know what all the groups are working on at any given point in time. If many groups are stuck in a similar place, I may interrupt the group work to have a whole class discussion. This allows for cross-fertilization of ideas between groups.

The total size of the class makes a big difference in how well the group work goes. A class of 18 is ideal. As the class size gets larger, it gets harder to get around to all the groups fast enough. One of the goals of the class is to have the students have a positive experience with group work and with discovery learning; in order to achieve this, the instructor needs to be ready to intervene whenever a group is stuck and isn't being productive. If the class is too large, it is harder to do this. I have, however, successfully taught this class to a group as large as 35 students. On the other hand, if the class gets smaller than 18, there is a danger of spending too much time with groups and not allowing them to work on their own. If this starts happening, I will sometimes just leave the classroom for five minutes at a time and go do other things, to give the groups enough time and space to work on their own.

The physical layout of the classroom also makes a big difference in how the class goes. Ideally, the whole course would be taught in a classroom with tables, with enough tables that each group can be by itself at a table. Even better is if the classroom also has computers around the outside for the computer labs and so that groups that are writing up their work together can work on the computers.

In interacting with students in groups and in class discussions, I try to always treat them as the experts and myself as a helpful questioner. So, for example, I always let groups fully explain to me what they are doing as though I have never seen their approach before, even if their approach is one that I have seen many times before. I try not to let them know from my affect whether or not I think their approach is correct until they have explained it to me. This creates an environment in which students take responsibility for explaining their ideas and expect to have them thoughtfully evaluated by themselves, their peers, and the instructor.

After we have spent a reasonable amount of time working on an assignment in class, writing it up will be assigned as homework. Since not all groups will work at the same pace, they won't all be in the same place. Often, the fastest groups will be able to start writing their answers up in class, while the slowest groups may or may not finish answering all the parts of the assignment. Usually I have some minimum core of the assignment in mind, and we don't stop working on it until all the groups have completed this core.

The write-ups are sometimes done in groups, and sometimes individually. Since this is the second semester I have had most of these students in this kind of class, I trust them to work well in groups and contribute equally, and therefore I assign more group assignments than I do during the first semester.

When informal assignments are submitted, I make a lot of comments on them, and grade them using a "circle" grading method in which each assignment is assigned a grade that is part of a circle. The percentage of the circle that is filled in corresponds to how close the assignment is to being completely correct. I then allow the students to respond to my comments and resubmit their assignments as many times as they would like to. Each time they resubmit an assignment, their grade can improve. The object of this grading system is to get students to focus on fixing their mistakes, and to (only) reward real progress made on an assignment. I also officially have the rule that students must obtain 3/4 of a circle within two weeks of the first draft being handed back, or else further progress after this point only gets half credit. This rule is there to make sure that students don't leave revisions to be done too long. I don't enforce the rule strictly if a student is making a reasonable effort to get revisions done and just hasn't gotten far enough.

Formal assignments are graded on both correctness and clarity of explanation. I write comments on the first draft, and then grade (but don't write many comments on) the second draft. I often allow students to submit a third draft if they haven't done well on the second draft provided that they come to my office hours to discuss the changes that need to be made first.

For both kinds of assignments, when I get tired of writing the same comments over and over again on student's papers, I compile lists of comments that I frequently make, type them up, and give them to the students, so that I can refer to them on papers just by writing a boxed code. Students don't take the boxed codes quite as seriously as they do handwritten comments, but it saves a lot of writing. I have included such comment sheets for some of the assignments in this course in Appendix A.

I generally try to keep the following materials on hand in the classroom so that they will be available to students if they need them:

- A classroom set of plastic Lénárt spheres, for when we are working on spherical geometry;
- Several crocheted hyperbolic planes, to show students before they make their own, and for students to use if they want them on a day they haven't brought their own;
- Paper, scissors, tape, and string, for making models of planes and cones and for making ellipses;
- Graph paper, for doing taxicab geometry;
- Geoboards, for our first day activity; and
- A set of Polydrons, for the activity on regular tessellations.

None of these are absolutely necessary, but they're all nice to have.

Several of the assignments use various computer packages. Ideally, you would be teaching this class in a room with computers around the outside that can be used whenever they are needed. Otherwise, you could try to meet in a computer lab on days when you know the computers will be needed, but note that once students become used to using computer tools to visualize geometric spaces, they often want to use them throughout the rest of the course. Now that laptops have become quite common, students may bring them to class to use, even if you don't have computers in your classroom.

Some of the assignments make use of Geometer's Sketchpad, but if you don't have access to Sketchpad, you could easily use any other dynamic geometry software such as Cabri or GeoGebra. GeoGebra is free and available for download from the website <http://www.geogebra.org/>. I find that students can generally complete these assignments using Sketchpad in computer labs, but some students prefer to have their own copies. I therefore usually ask the campus store to have the student version of Sketchpad available as an optional course material. The other computer programs we use, Spherical Sketchpad, non-Euclid, and KaleodoTile, are all available for free on the web.

First Day of Class

On the first day of class, after we discuss the syllabus, we usually work on an activity that I don't collect, whose purpose is to get the students back into the problem-solving state of

mind required for this course. The question we look at is: how many different kinds of non-congruent squares can be made with rubber bands on a 5x5 geoboard?

Students work on this activity in groups, and generally have several a-ha moments, as they realize that there are squares that aren't horizontal, and then that there are squares that can be made with two or more rubber bands that can't be made with a single rubber band (whose vertices don't lie on pegs) . Hopefully, eventually some students start trying to come up with a systematic way to find all of the squares.

We then relate their experiences doing this activity to the Van Hiele levels, and I give them some information about the Van Hiele levels taken from [Dossey et al., 2002, pp. 203–204]. I ask them to think about these levels throughout the semester.

I first learned about this activity from Jodie Novak, but I don't know where it originated.

Chapter 1

Formal Assignment 1: Taxicab Geometry

Write a group paper about taxicab geometry. In your paper, you should explain what Taxicab geometry is and what you have learned about it. You can look at some of the questions below, or any other questions that seem interesting to you.

Some possible questions:

- Can you give a formula for taxicab distance? Is taxicab distance always less or more than Euclidean distance? Can you prove your answer?
- What do taxicab circles look like? How can they be used? What is the taxicab value of π ?
- What do taxicab ellipses look like? How can they be used? (Recall, an ellipse with foci A and B is $\{P \mid d(A, P) + d(B, P) = c\}$, the set of all of the points that can be drawn using a string of length c attached to points A and B .)
- What happens if we look instead at a set $\{P \mid d(A, P) - d(B, P) = c\}$ in Euclidean or taxicab geometry? What happens in the case where $c = 0$?
- How do you find the distance from a point to a line in taxicab geometry? Can you give a careful procedure that always works? How does this differ from euclidean geometry? If I have a point A and a line L , what is $\{P \mid d(A, P) = d(P, L)\}$ in Euclidean or taxicab geometry?
- Can you use Geometer's Sketchpad to model any of these shapes in taxicab geometry?
- Are there any theorems of Euclidean geometry that aren't true in taxicab geometry?

Teaching Notes:

We typically spend two weeks on this project. I assign it as a group formal assignment, and try to assign groups that mix together students who took the first semester course at different times. The assignment eases them back into working in groups and exploring a new geometry. It gives the students an opportunity to think about the definitions of some shapes that they probably haven't thought about much before.

Students can start working on this project immediately without much set up. I explain to them that in taxicab geometry, distances on the plane are measured as they would be to a taxicab on a grid of city streets that can only travel north-south and east-west. I give a simple example, and then set them working on the above questions. I purposely do not give

them a written definition of taxicab geometry, since I want students to come up with their own explanation for their papers.

This assignment also gives us a chance to get back into working with Geometer's Sketchpad. After they have been working on this project for a day or two, I show the students how to construct ellipses in Sketchpad as the locus of the intersection points of two circles constructed so that the sum of their radii is constant. I also show them how to construct taxicab circles on Sketchpad. Many groups are then able to use these ideas to construct taxicab ellipses and hyperbolas using Sketchpad.

The length of this project gives students a chance to work on building their hyperbolic planes, as discussed in the next section. It also helps them to understand the Poincaré disc later on, since that is another geometry with a non-standard metric.

Chapter 2

Informal Assignment 1: Build your own Hyperbolic Plane!

Build your own hyperbolic Plane! You can use any one of the four methods explained in appendix B of *Experiencing Geometry*. The crocheted hyperbolic planes are by far the nicest, but the hardest to make. This assignment will be graded on the quality of your finished product.

Teaching Notes:

I give the students this assignment and talk about it well in advance of when it will be due. I try to give them two weeks to work on it. I am always surprised at just how many people are willing to learn how to crochet in order to do this assignment. It is also helpful if someone has used the paper annuli method to create a hyperbolic plane, since seeing that construction helps everyone understand how the crocheted hyperbolic planes work, and it also is referred to in Problem 5.1.

Chapter 3

Informal Assignment 2: Straightness and Euclid's Postulates on the Hyperbolic Plane

Decide which lines are straight on your Hyperbolic Plane. In doing so, you may find it helpful to look at Problem 5.1. Which of Euclid's five Postulates are true on the Hyperbolic Plane? Also look at the modified version of his first postulate that we will call Postulate 1': Two points determine a unique line.

Your write-up of this assignment should include explanations of which lines are straight on the hyperbolic plane and how you know they are straight, and for each postulate, whether it is true or false, and how you know.

Teaching Notes:

Euclid's Postulates are given in Appendix B of these notes. They also appear in Appendix A of *Experiencing Geometry*. The difference between Postulate 1 and Postulate 1' is that Postulate 1 doesn't require the line between two given points to be unique.

This assignment asks students to consider the hyperbolic planes that they have just constructed for their previous assignment, to decide what lines are straight on this new surface, and then to decide which of Euclid's postulates are true. If students took the first semester version of this course, then they have each written a definition of straightness that they can use in evaluating which lines are straight on this surface. If just some of the students have done this, then they can fill everyone else in on possible definitions of straightness. Otherwise, if no one in the class has done this before, you would want to spend some time here discussing Problem 1.1.

This assignment poses essentially the same question that we looked at first with every new space we encountered in the first semester course. So if they took this course, the students should be getting better at deciding when postulates are true, but the hyperbolic plane will be significantly more confusing than any other surface they have looked at previously. I encourage students to use the questions from the textbook to help figure out what the straight lines are. We generally end up doing Problem 5.1a as a whole class discussion.

We also discuss the history of Euclid's Fifth Postulate and the history of the discovery of the hyperbolic plane, which are intertwined. Many people had tried to prove the fifth postulate from the first four, on the grounds that it seemed too complicated to be a postulate. The hyperbolic plane is a surface in which the first four postulates are true, but the fifth is not, so it shows the impossibility of proving the fifth postulate from the first four, since any such proof would have to work in the Hyperbolic Plane. In fact, the hyperbolic plane was discovered in the nineteenth century by mathematicians who were trying to prove the fifth

postulate from the first four. They tried to give a proof by obtaining a contradiction from the denial of the fifth postulate, but eventually discovered that they were just working out theorems of hyperbolic geometry.

When I get tired of writing the same comments over and over again on student's papers, I compile lists of comments that I frequently make, type them up, and give them to the students, so that I can refer to them on papers just by writing a boxed code. Students don't take the boxed codes quite as seriously as they do handwritten comments, but it saves a lot of writing. I have included such comment sheets for some of the assignments in this course, including this one, in Appendix A.

Chapter 4

Informal Assignment 3: Non-Euclid

Go to the webpage for the non-Euclid applet, which is found at <http://www.cs.unm.edu/~joel/NonEuclid/NonEuclid.html>.

This applet provides a dynamic geometry model of one kind of hyperbolic geometry, called the Poincaré Disk model. Notice that it is a kind of taxicab geometry: the distance between two points in the Poincaré disk is given by a non-Euclidean distance function.

Experiment with this model. It seems very different from the hyperbolic planes that we have been looking at. Do you think it is reasonable to call this a model of hyperbolic geometry? Why or why not? You may also find it useful to compare this model with the other model provided by non-Euclid, which is called the Upper Half-Plane model.

Try to come up with as many examples as you can of theorems of Euclidean geometry that are not true in the Poincaré Disk model, and of facts that appear to be true in the Poincaré Disk model that are not true in Euclidean geometry.

Teaching Notes: This provides a very different model of hyperbolic geometry, and having two different models significantly helps students in thinking about hyperbolic geometry. I don't introduce it as another model of the same space; rather, I let the students explore the space and gradually realize that it is intrinsically the same as the space they were just looking at.

One reason that students may give for this space being hyperbolic is that the fifth postulate is false here, just as it is in the hyperbolic plane, particularly if you have discussed the history of the Fifth Postulate and the hyperbolic plane.

Asking the students to come up with facts that are different between the Poincaré disc, the physical hyperbolic plane, and the Euclidean plane leads to very interesting discussions. Different definitions of parallel lines often arise naturally in this context. Students also often make discoveries about the sum of angles in a triangle, about what it means to try to make a square on this surface, and other interesting facts about the Hyperbolic Plane.

After we have discussed this as a class for a while and no one has been able to come up with any intrinsic differences between the hyperbolic plane and the Poincaré disc, I admit to them that the two spaces are intrinsically the same, and tell them that they can use them interchangeably.

Chapter 5

Informal Assignment 4: Folding a parallel line

5.1 Part 1

Assume that you have been given a piece of paper containing a straight line and a point not on that straight line. Figure out how to create a straight line through the point that is parallel to the original line, using only paper folding. Make sure that your method doesn't require any estimation on the part of the person doing the folding. Write up your method in a series of steps that someone else could follow.

5.2 Part 2

Prove that your method always works—that is, that the line created by your method can never intersect the original line. Your argument should not include any facts that you don't know how to prove. You will probably find Problem 8.2 helpful here.

Teaching Notes:

I give this assignment as a way of leading into our discussion of parallel lines. I initially just assign part 1. On the day it is due, we discuss it, but I don't collect it. Instead, I extend it by adding part 2, and we work on Problem 8.2 in groups in class. (See Appendix C for the statement of Problem 8.2 and the other problems discussed in these notes.) The students must eventually write up both parts together, and submit them as an individual informal assignment.

Students' methods of creating a parallel line almost invariably involve creating parallel transports. (In fact, most of the time they create two lines with a common perpendicular.) By Problem 8.2, their figure then has half-turn symmetry, so that if the lines intersected on one side, they would intersect on the other, which is impossible on the plane.

Problem 8.2 is generally quite hard for students since they are unlikely to have seen symmetry proofs before. Usually, at some point while they are working on this problem, we discuss symmetry proofs as a class, and, in particular, we discuss a symmetry proof of the Isosceles Triangle Theorem. The proof goes as follows: if we are given a triangle ABC in which AB is congruent to AC , then if we form the angle bisector of angle A and call it L , then the triangle has reflection symmetry across L . This is because if we reflect across L , ray AB switches places with ray AC , since L is the angle bisector; B switches places with C , because they are the same distance along these rays, so segment AB switches places with

segment AC ; and line segment BC stays in place since B and C switch places and these two points determine a unique line segment in the plane. So the triangle has reflection symmetry over L , and so angles B and C must therefore switch places, so they must be congruent to one another.

Even once we have gone over Problem 8.2, students have trouble putting the whole argument together. In fact, after I've given back the first draft of this assignment, we usually have a class discussion laying out the whole argument for why the original lines must be parallel, but students still have trouble afterwards writing the whole thing down correctly. It's a really good exercise for them, though.

5.3 Informal Assignment 5: Problems 8.3 and 8.4

After we work on Problem 8.2, we continue on to Problems 8.3 and 8.4. (See Appendix C for the statements of these problems.) In addition to being central to our study of parallel lines, these problems nicely incorporate all of the surfaces we have looked at (even including the 450° cone!). They also lead into the questions asked on the midterm. For more on the midterm, see Appendix E. I view this collection of problems (8.2, 8.3, 8.4, and the related midterm questions) as being the central core material of the course.

A useful resource for students as they are working on these problems is Spherical Easel. Spherical Easel is a free java application created by David Austin and Will Dickinson, and is available at <http://merganser.math.gvsu.edu/easel/>. Students who have already had experiences with Geometer's Sketchpad and non-Euclid can probably figure out how to use Spherical Easel on their own. Note that once they have been introduced to Spherical Easel, the students have access to both physical and computer models of the plane, of the sphere, and of the hyperbolic plane.

5.4 In-class Discussion: Problem 7.2

I often also spend a class period going over Problem 7.2, which shows how find the area of a triangle on a hyperbolic plane. This problem is difficult enough that I don't generally have students do it on their own or turn it in, but it is nice to do as a class because

- the proof is exactly analogous to the proof of Problem 7.1 on finding the area of a spherical triangle, which they did the semester before;
- it explains why the angle sum of a triangle on the hyperbolic plane is less than 180° ;
- constructing the diagrams for the proof is a cool use of the non-Euclid applet; and
- using these diagrams along with a Smart Board to show the proof is even cooler.

Chapter 6

Regular Tessellations and Polyhedra

You have been given a set of polydrons (snap-together polygons) to work with. We are going to use the polydrons to produce **regular** figures—that is, figures where the same number of a given regular polygon meet at each vertex. Using your polydrons, complete the following table. If a given combination produces a planar tessellation, draw it; if it produces a polyhedron, name it according to how many faces it has.

# of shapes meeting at each vertex	3	4	5	6	7
Triangles					
Squares					
Pentagons					
Hexagons					
Heptagons					

The following list of greek numerical prefixes may be useful in naming the polyhedra:

1	mono
2	di
3	tri
4	tetra
5	penta
6	hexa
7	hepta
8	octa
9	ennea
10	deca
11	hendeca
12	dodeca
13	triskaideca
14	tetrakaideca
15	pentakaideca
16	hexakaideca
17	heptakaideca
18	octakaideca
19	enneakaideca
20	icosa
24	icositetra
30	triconda or triaconta
40	tetraconta
50	pentaconta

Teaching Notes:

This is an activity we do in class one day as a break. I don't collect it, and students generally have a lot of fun with it. Polydrons can be purchased on the web or from many educational supply companies.

The correct answer is:

# of shapes meeting at each vertex	3	4	5	6	7
Triangles	Tetrahedron	Octahedron	Icosahedron	Planar Triangular Tessellation	Hyperbolic Tessellation
Squares	Cube	Bathroom Tile Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation
Pentagons	Dodecahedron	Hyperbolic Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation
Hexagons	Honeycomb Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation
Heptagons	Hyperbolic Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation	Hyperbolic Tessellation

Students will need some guidance with the hyperbolic tessellations, since they have a tendency to try to make them join up in the middle rather than just expanding out forever.

Some instructor intervention while they are working on these can help keep them from spending too long going in the wrong direction.

Note that 7 triangles meeting at a vertex gives one of the possible constructions of (an approximation to) the hyperbolic plane discussed in Appendix B of *Experiencing Geometry*. This approximation by triangles can actually be projected onto a (not-approximated) tessellation of the hyperbolic plane by hyperbolic equilateral triangles with interior angles of $360/7^\circ$. Something similar happens in all the other boxes labeled as hyperbolic tessellations. In much the same way, all of the polyhedra can be projected onto tessellations of the sphere in which the angles expand, rather than shrink. Nice illustrations of these can be given with the program KaleidoTile. KaleidoTile is a free computer program written by Jeff Weeks, and is available at

<http://www.geometrygames.org/KaleidoTile/index.html>.

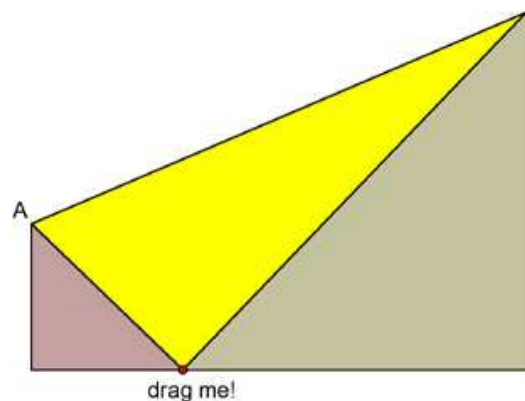
I generally take one other day towards the end of the semester to show the students KaleidoTile.

The table also shows why there are three regular tessellations of the plane and five regular polyhedra (the Platonic solids).

Chapter 7

Computer Lab I: Paper folding

When the top left corner of a piece of paper is folded down to touch the bottom edge of the paper, a triangle is formed in the lower left corner, as shown below. Model this situation in Sketchpad. Now use the sketch to answer the following question: When does the lower-left triangle have the biggest possible area?



Further challenges:

1. Can you prove that your answer is correct? It may be helpful to recall that you probably learned about maximization problems in a calculus class.
2. Your sketch probably falls apart when the point is dragged too close to the lower-right corner. Extend your sketch to show (and smoothly transition into) this third possible state. Can you construct a sketch in which the corner point can be dragged all the way around the edge of the piece of paper?

What to turn in:

- Turn in a sketchpad file electronically by emailing it to your instructor. See how pretty you can make it! The file you submit should be named by your last name(s).
- Also turn in a written explanation of when the triangle is as big as possible, and your proof that you're right.

Teaching Notes:

I normally collect the electronic file as informal assignment 6, and the written explanation as informal assignment 7.

This assignment is a slightly modified version of Project 64 from the booklet *101 Project Ideas for The Geometer's Sketchpad*, which comes with Geometer's Sketchpad. It is a nice combination of geometry, modeling, and calculus. We work on it as a change of pace while the students are working on the midterm at home.

One really nice thing to do with this activity is to put the model of the paper onto a coordinate system and use Sketchpad to graph the area of the triangle as a function of the distance of the *drag me* point from the lower left-hand corner of the paper. The maximum of this graph corresponds to the point we are looking for.

Chapter 8

Computer Lab II: The Product of Two Reflections

Examine what happens when you reflect a figure consecutively over two separate lines. What is the resulting motion of the figure? State your findings as precisely as possible, and see if you can prove them. Be sure to account for all possible arrangements of the lines.

Teaching Notes:

I normally collect this assignment as informal assignment 8.

Much more structured versions of this activity are often included in secondary level curricula. (For example, the activities “Slide Me Now” and “Design This” from the NCTM Navigations series [Day et al., 2001]; “Investigation: Reflections Across Intersecting Lines” and “Investigation: Reflections Across Parallel Lines” from *Exploring Geometry with the Geometer’s Sketchpad* [Bennett, 1996]; Problem 2.4 from *Kaleidoscopes, Hubcaps, and Mirrors* from the *Connected Mathematics* eighth grade curriculum [Lappan et al., 2004]; and Unit 2, Lesson 2, Investigation 3 from the tenth grade *Contemporary Mathematics in Context* curriculum [Coxford et al., 2003], to name just a few.) But when posed in this open-ended way, it makes a perfect investigation for pre-service secondary teachers as well. I am more or less lenient about how many cases the proofs must include depending on how strong the students are and how fast I want to get through this assignment.

Chapter 9

Informal Assignment 9: Parallel Postulates

An excerpt from the book *Gödel, Escher, Bach: an Eternal Golden Braid* [Hofstadter, 1999] is reproduced in Appendix D of these notes. On the last page of this excerpt, Douglas Hofstadter claims, without explanation, that Euclid's Fifth Postulate (EFP) and the Parallel Postulate (PP) are "equivalent." However, he doesn't say what he means by the word "equivalent" in this context. Show that this definition matters by giving a careful definition of the word "equivalent" under which these statements are not equivalent, and explain why; then give a (different) careful definition of this word under which these statements are equivalent, and prove that they are in fact equivalent under your definition.

In answering this question, you may find it helpful to read *Experiencing Geometry* Problem 10.3; the assumption that Hofstadter calls the Parallel Postulate is equivalent (in both senses!) to what *Experiencing Geometry* calls the High School Parallel Postulate (HSP), because it is commonly assumed in high school geometry textbooks.

Teaching Notes:

Before we begin working on this assignment I give the students the excerpt reproduced in Appendix D from *Gödel, Escher, Bach* [Hofstadter, 1999] as a reading assignment.

EFP and PP are not equivalent in that they are not true on exactly the same surfaces. EFP is true on the sphere, while PP is not. Once students realize this, they usually conclude that Hofstadter is incorrect in his use of the word "equivalent." The two postulates are, however, equivalent in the sense that either one can be proven from the other along with all the other postulates (including 1'). Another way of saying this is that they are true on exactly the same surfaces, as long as we restrict ourselves to surfaces on which all the other postulates are true.

I usually assign this as an individual informal assignment. It allows the students to think about the meaning of the word "equivalent" in mathematics, and deepens ideas introduced on the midterm, where they also had to prove that two statements were equivalent in the sense that each could be proven from the other.

Chapter 10

Final Projects

For your final project, pick a geometric topic that we have not covered in either semester of this course and explore it on your own.

Your project will consist of three parts:

1. A written paper;
2. A ten minute in-class presentation; and
3. A model of some kind related to your topic. Your model can be a physical model or a computer model.

You can look at any topic, as long as I approve it. Many of my suggested topics come from parts of the *Experiencing Geometry* textbook that we have not studied previously. Each project must include some proof(s) of some kind. People working on the same or related topics may work together, but a group of two people will be expected to cover twice as much material as a single person working alone.

Some possible topics:

- Triangle congruence theorems: ASA (Problem 6.5), SSS (9.1), ASS (9.2), SAA (9.3), AAA (9.4)
- Parallel Postulates: 10.2, 10.3, 10.4
- Patterns: 11.5, 11.6
- Spherical Projections and map making: 14.1 14.4. (14.1 + 14.2 together count as one problem, and only for a group also doing 14.3 and/or 14.4)
- Upper half-plane and Poincaré Disk models: 17.1-17.6
- 2-Manifolds (Flat torus, Klein bottle, etc.): (18.1+18.2) (flat), 18.3 (spherical), 18.4 (hyperbolic), 18.5 (Euler Number), 18.6 (2D Shape of Space)
- Dissection Theory: Ch. 12, 13.
- Geometric algebra; Euclid's *Elements* Book II; and Geometric solutions of Quadratic and Cubic Equations (19.1-19.4)

- Trigonometry and Duality: 20.1 (intrinsic circumference of a circle), 20.2 (spherical law of cosines), 20.3 (spherical law of sines), (20.4 + 20.5)(Duality and the laws of sines/cosines) or (20.4 + 20.6) (Duality + Projective plane) (relates to perspective drawing and 18.3)
- Mechanisms 21.1, 21.2 (Four bar linkages); 21.3 (Reuleaux triangles)
- 4D space: 22.1, only if in a group doing other problems from this chapter; 22.2 (3-Spheres), 22.3-22.6
- Polyhedra: 23.1-23.4 (Spherical angles and tetrahedra congruence theorems)
- Demi-regular tessellations, symmetries of tessellations, etc.
- Tessellations of $\mathbf{R}^3$ by polyhedra (This topic is related to chemistry and how crystals are formed.)
- Questions related to astronomy: astrolabe, calendars, sundial, orrery
- Golden Ratio

Teaching Notes:

This assignment lets the students show that they are now able to explore new areas on their own, and lets them see a wide range of further applications of geometry all at once. For those reasons it is a good ending activity. Most of these topics are taken from sections of the book that we haven't covered. I have the students present their work at our final meeting time. It is sometimes an issue having enough time for everyone to present, because students consistently underestimate the amount of time their presentations will take.

Appendix A

Comment Sheets

The following sections contain comments that are frequently made on papers in this course. If you get a paper back with a comment that is just a letter and number in a box, look here to find the corresponding comment.

A.1 Hyperbolic Plane Comments

H1. How do you know that radial lines have the symmetries of a straight line? Problem 5.1 suggests that you use the local intrinsic symmetries of the annular strips. Which symmetry of straight lines do the radial lines clearly have on a single annular strip?

H2. Good. What property of straight lines does this relate to?

H3. The easiest way to show that all right angles, at all points on the hyperbolic plane, are equal to one another is to show that all angles that meet Euclid's definition of being right are 90 degrees, and therefore they're all equal.

H4. Can you give a specific counterexample for Euclid's fifth postulate?

H5. Ok—but how do you know that a straight line can be drawn connecting any two points? Can you describe how you would actually do it given two points on a physical hyperbolic plane?

H6. The word “parallel” normally is defined to mean “non-intersecting.”

H7. There are lines that intersect on the hyperbolic plane, as well as lines that don't intersect and don't get close together. The asymptotic lines are very special in that they get arbitrarily close together, but don't intersect.

H8. How do you know that the radial lines can never intersect?

H9. How does this help show that Euclid's Fifth postulate is false?

A.2 Folding a Parallel Line Comments

A1. You need to carefully show that each part of your figure lands on another part of the figure when rotated 180 degrees about the center point, like we did in class. Start from the center and work your way outward.

A2. How do you know that this always creates a perpendicular line? Can you prove that the lines are perpendicular?

A3. The word “parallel,” used without any qualification, is normally understood to mean “non-intersecting.” Is that what you mean here?

A4. How do you know that the figure that you folded produces parallel transports?

A5. Why does this mean that the two lines can’t ever intersect?

A.3 Parallel Postulates Comments

B1. Did you start with the right beginning? Remember, if you’re trying to prove a statement B by assuming statement A , and if B is an *if-then* statement, then you need to assume the *if* part of B is true and demonstrate in that case that the *then* part of B is true. Somewhere in this argument you will have to use the fact that A is true.

B2. If you’re showing that two statements can be used to prove one another, you can’t assume that you’re working in the plane. Statements are simply true or false in the plane. Proofs, on the other hand, take place in an assumed axiom system, and therefore deal with all possible models of the assumed axioms.

B3. What, exactly, are the two different definitions of the word “equivalent” that you are working with? You need to explicitly state them.

B4. You’ve shown there is at most one line through the point parallel to the original line. But how do you know that there are any lines through the point that are parallel to the original line?

B5. What is the sense in which the two postulates are not equivalent?

B6. Why must this line be parallel to your original line?

B7. Good! Why does your proof that EFP and PP can be used to prove one another show that the same conclusions can be drawn from each of them?

Appendix B

From Book I of Euclid's *Elements*

Euclid wrote the thirteen books of the *Elements* around 300 B.C. The first four books are about elementary planar geometry; books five through ten are largely about the theory of ratio and proportion and about what would now be considered number theory; and books eleven through thirteen are about three dimensional geometry.

The excerpt reprinted here is taken from the edition translated and edited by Thomas Heath [Euclid, 1956].

B.1 Definitions

1. A **point** is that which has no part.
2. A **line** is breadthless length.
3. The extremities of a line are points.
4. A **straight** line is a line which lies evenly with the points on itself.
5. A **surface** is that which has length and breadth only.
6. The extremities of a surface are lines.
7. A **plane** surface is a surface which lies evenly with the straight lines on itself.
8. A **plane angle** is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.
9. And when the lines containing the angle are straight, the angle is called **rectilineal**.
10. When a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is **right**, and the straight line standing on the other is called a **perpendicular** to that on which it stands.
11. An **obtuse** angle is an angle greater than a right angle.
12. An **acute** angle is an angle less than a right angle.
13. A **boundary** is that which is an extremity of anything.
14. A **figure** is that which is contained by any boundary or boundaries.

15. A **circle** is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure are equal to one another;
16. And the point is called the **centre** of the circle.
17. A **diameter** of the circle is any straight line drawn through the centre and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.
18. A **semicircle** is the figure contained by the diameter and the circumference cut off by it. And the centre of the semicircle is the same as that of the circle.
19. **Rectilineal figures** are those which are contained by straight lines, **trilateral** figures being those contained by three, **quadrilateral** those contained by four, and **multilateral** those contained by more than four straight lines.
20. Of trilateral figures, an **equilateral** triangle is that which has its three sides equal, an **isosceles** triangle that which has two of its sides alone equal, and a **scalene** triangle that which has its three sides unequal.
21. Further, of trilateral figures, a **right-angled** triangle is that which has a right angle, an **obtuse-angled** triangle that which has an obtuse angle, and an **acute-angled** triangle that which has its three angles acute.
22. Of quadrilateral figures, a **square** is that which is both equilateral and right-angled; an **oblong** [rectangle] that which is right-angled but not equilateral; a **rhombus** that which is equilateral but not right-angled; and a **rhomboid** [parallelogram] that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called **trapezia**.
23. **Parallel** straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

B.2 Postulates

Let the following be postulated:

1. To draw a straight line from any point to any point.
2. To produce a finite straight line continuously in a straight line.
3. To describe a circle with any centre and distance.
4. That all right angles are equal to one another.
5. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

B.3 Common Notions

1. Things which are equal to the same thing are also equal to one another.
2. If equals be added to equals, the wholes are equal.
3. If equals be subtracted from equals, the remainders are equal.
4. Things which coincide with one another are equal to one another.
5. The whole is greater than the part.

Appendix C

Problems from *Experiencing Geometry*

These problems are taken from *Experiencing Geometry: Euclidean and non-Euclidean with History* [Henderson & Tamiņa, 2005].

Problem 1.1: When Do You Call a Line Straight?

- a. How can you check in a practical way if something is straight? How do you construct something straight—lay out fence posts in a straight line, or draw a straight line? Do this without assuming that you have a ruler, for then we will ask, “How can you check that the ruler is straight?”
- b. What symmetries does a straight line have?
- c. What is in common among the different notions of straightness? Can you write a definition of “straight line”?

Problem 2.1: What is Straight on a Sphere?

- a. Imagine yourself to be a bug crawling around on a sphere. (This bug can neither fly nor burrow into the sphere.) The bug’s universe is just the surface; it never leaves it. What is “straight” for this bug? What will the bug see or experience as straight? How can you convince yourself of this? Use the properties of straightness (such as symmetries) that you talked about in Problem 1.1.

Problem 5.1: What is Straight in a Hyperbolic Plane?

- a. On a hyperbolic plane, consider the curves that run radially across each annular strip. Argue that these curves are intrinsically straight. Also, show that any two of them are asymptotic, in the sense that they converge toward each other but do not intersect.
- b. Find other geodesics on your physical hyperbolic surface. Use the properties of straightness (such as symmetries) you talked about in Problems 1.1, 2.1, and 4.1.
- c. What properties do you notice for geodesics on a hyperbolic plane? How are they the same as geodesics on the plane or spheres, and how are they different from geodesics on the plane and spheres?

Problem 7.1: The Area of a Triangle on a Sphere

- The two sides of each interior angle of a triangle Δ on a sphere determine two congruent lunes with lune angle the same as the interior angle. Show how the three pairs of lunes determined by the three interior angles α , β , γ , cover the sphere with some overlap. (What is the overlap?)
- Find a formula for the area of a lune with lune angle θ in terms of θ and the (surface) area of the sphere (of radius ρ), which you can call S_ρ . Use radian measure for angles.
- Find a formula for the area of a triangle on a sphere of radius ρ .

Problem 7.2: Area of Hyperbolic Triangles

- On your hyperbolic plane draw as large a triangle as you can find. Compare your triangle with the large triangles that others have found. What do you notice?
- Show that on the same hyperbolic plane, all $2/3$ -ideal triangles with the same angle θ are congruent. [A $2/3$ -ideal triangle is one with two vertices at infinity.]
- Show that the area function A_ρ is an additive function. That is,

$$A_\rho(\alpha + \beta) = A_\rho(\alpha) + A_\rho(\beta).$$

[Here, $A_\rho(\alpha)$ denotes the area of a $2/3$ -ideal triangle with exterior angle α on a hyperbolic plane of radius ρ .]

- Find a formula for the area of a hyperbolic triangle.

Problem 8.2: Symmetries of Parallel Transported Lines

Consider two lines, r and r' , that are parallel transports of each other along a third line, l . [Two lines r and r' are said to be parallel transports along a transversal l if l cuts r and r' at congruent angles.] Consider now the geometric figure that is formed by the three lines, one of them being a transversal to the other two, and look for the symmetries of that geometric figure.

What can you say about the lines r and r' ? Do they intersect? If so, where? Look at the plane, spheres, and hyperbolic planes.

Problem 8.3: Transversals through a Midpoint

- Prove: If two geodesics r and r' are parallel transports along another geodesic l , then they are also parallel transports along any geodesic passing through the midpoint of the segments between r and r' . Does this hold for the plane, spheres, and hyperbolic planes?
- On a sphere or hyperbolic plane, are these the only lines that will cut r and r' at congruent angles? Why?

- c. Prove: Two geodesics (on the plane, spheres, or hyperbolic planes) are parallel transports of each other if and only if they have a common perpendicular. On spheres and hyperbolic planes the common perpendicular is unique.

Problem 8.4: What is “Parallel”?

...[Consider] these common definitions for parallel lines in the plane:

1. Parallel lines are lines that never intersect;
 2. Parallel lines are lines such that any transversal cuts them at congruent angles; or
 3. Parallel lines are lines that are everywhere equidistant.
- a. Check out for each of these three definitions whether they apply to parallel transported lines on a sphere or on a hyperbolic plane.
- b. Show that there are pairs of geodesics on a hyperbolic plane that do not intersect and yet there are **no** transversals that cut at congruent angles. That is, the geodesics are parallel (in the sense of not intersecting) but not parallel transports of each other.
- c. Show that there are pairs of geodesics on any cone with cone angle greater than 360° that do not intersect and yet there are **no** transversals that cut at congruent angles. That is, the geodesics are parallel (in the sense of not intersecting) but not parallel transports of each other along any straight (in the sense of symmetry) line.

Teaching Notes: Several problems from *Experiencing Geometry: Euclidean and non-Euclidean with History* [Henderson & Tamiya, 2005] discussed in this course are reproduced here. In the original book, each problem is given along with extensive commentary which students need to read in order to make sense of the problems. I normally ask students to purchase this book, and the brief statements of the problems here are not meant to replace the book. However, if you were going to just use some of these materials, you could put together a course packet reproducing the relevant sections of the book. Most campus stores can arrange to obtain the rights to reproduce an excerpt of a book in a course packet.

Appendix D

Excerpt from *Gödel, Escher, Bach*

The following excerpt is taken from Chapter 4 of *Gödel, Escher, Bach: an Eternal Golden Braid* by Douglas Hofstadter [Hofstadter, 1999, pp. 88–93]. This book won the Pulitzer Prize for general non-fiction in 1980.

Prior to this excerpt, Hofstadter has been discussing a formal system that he calls the pq-system which allows you to derive certain strings of symbols. An example of a statement provable in the pq-system is

$$- - - p - - q - - - - - .$$

Hofstadter points out that all of the statements in the pq-system will correspond to true statements of arithmetic if we interpret a statement like this as meaning “three plus two equals five”. However, he then points out that they are equally all true if we interpret it as meaning “three equals two subtracted from five” or “three plus two is less than or equal to five”. Thus, there are multiple possible meaningful interpretations of the statements of the formal system.

The excerpt follows.

The History of Euclidean Geometry

Although I have tried to catch you off guard and surprise you a little, this lesson about how to interpret symbols by words may not seem terribly difficult once you have the hang of it. In fact, it is not. And yet it is one of the deepest lessons of all of nineteenth century mathematics! It all begins with Euclid, who, around 300 B.C., compiled and systematized all of what was known about plane and solid geometry in his day. The resulting work, Euclid’s *Elements*, was so solid that it was virtually a bible of geometry for over two thousand years—one of the most enduring works of all time. Why was this so?

The principal reason was that Euclid was the founder of rigor in mathematics. The *Elements* began with very simple concepts, definitions, and so forth, and gradually built up a vast body of results organized in such a way that any given result depended only on foregoing results. Thus, there was a definite plan to the work, an architecture which made it strong and sturdy.

Nevertheless, the architecture was of a different type from that of, say, a skyscraper. . . . In the latter, that it is standing is proof enough that its structural elements are holding it up. But in a book on geometry, when each proposition is claimed to follow logically from earlier propositions, there will be no visible crash if one of the proofs is invalid. The

girders and struts are not physical, but abstract. In fact, in Euclid's *Elements*, the stuff out of which proofs were constructed was human language—that elusive, tricky medium of communication with so many hidden pitfalls. What, then, of the architectural strength of the *Elements*? Is it certain that it is held up by solid structural elements, or could it have structural weaknesses?

Every word which we use has a meaning to us, which guides us in our use of it. The more common the word, the more associations we have with it, and the more deeply rooted is its meaning. Therefore, when someone gives a definition for a common word in the hopes that we will abide by that definition, it is a foregone conclusion that we will not do so but will instead be guided, largely unconsciously, by what our minds find in their associative stores. I mention this because it is the sort of problem which Euclid created in his *Elements*, by attempting to give definitions of ordinary, common words such as “point”, “straight line”, “circle”, and so forth. How can you define something of which everyone already has a clear concept? The only way is if you can make it clear that your word is supposed to be a technical term, and is not to be confused with the everyday word with the same spelling. You have to stress that the connection with the everyday word is only suggestive. Well, Euclid did not do this, because he felt that the points and lines of his *Elements* were indeed *the* points and lines of the real world. So by not making sure that all associations were dispelled, Euclid was inviting readers to let their powers of association run free . . .

This sounds almost anarchic, and is a little unfair to Euclid. He did set down axioms, or postulates, which were supposed to be used in the proofs of propositions. In fact, nothing other than those axioms and postulates was supposed to be used. But this is where he slipped up, for an inevitable consequence of his using ordinary words was that some of the images conjured up by those words crept into the proofs which he created. However, if you read proofs in the *Elements*, do not by any means expect to find glaring “jumps” in the reasoning. On the contrary, they are very subtle, for Euclid was a penetrating thinker, and would not have made any simple-minded errors. Nonetheless, gaps are there, creating slight imperfections in a classic work. But this is not to be complained about. One should merely gain an appreciation for the difference between absolute rigor and relative rigor. In the long run, Euclid's lack of absolute rigor was the cause of some of the most fertile path-breaking in mathematics, over two thousand years after he wrote his work.

Euclid gave five postulates to be used as the “ground story” of the infinite skyscraper of geometry, of which his *Elements* constituted only the first several hundred stories. The first four postulates are rather terse and elegant:

1. A straight line segment can be drawn joining any two points.
2. Any straight line segment can be extended indefinitely in a straight line.
3. Given any straight line segment, a circle can be drawn having the segment as radius and one end point as center.
4. All right angles are congruent.

The fifth, however, did not share their grace:

5. If two lines are drawn which intersect a third in such a way that the sum of the inner angles on one side is less than two right angles, then the two lines inevitably must intersect each other on that side if extended far enough.

Though he never explicitly said so, Euclid considered this postulate to be somehow inferior to the others, since he managed to avoid using it in the proofs of the first twenty-eight

propositions. Thus, the first twenty-eight propositions belong to what might be called “four-postulate geometry”—that part of geometry which can be derived on the basis of the first four postulates of the *Elements*, without the help of the fifth postulate. (It is also often called *absolute geometry*.) Certainly Euclid would have found it far preferable to *prove* this ugly duckling, rather than to have to *assume* it. But he found no proof, and therefore adopted it.

But the disciples of Euclid were no happier about having to assume this fifth postulate. Over the centuries, untold numbers of people gave untold years of their lives in attempting to prove that the fifth postulate was itself part of four-postulate geometry. By 1763, at least twenty-eight different proofs had been published—all erroneous! (They were all criticized in the dissertation of one G. S. Klügel.) All of these erroneous proofs involved a confusion between everyday intuition and strictly formal properties. It is safe to say that today, hardly any of these “proofs” holds any mathematical or historical interest—but there are certain exceptions.

The Many Faces of Noneuclid

Girolamo Saccheri (1667–1733) lived around Bach’s time. He had the ambition to free Euclid of every flaw. Based on some earlier work he had done in logic, he decided to try a novel approach to the proof of the famous fifth: suppose you *assume its opposite*; then work with *that* as your fifth postulate . . . Surely after a while you will create a contradiction. Since no mathematical system can support a contradiction, you will have shown the unsoundness of your own fifth postulate, and therefore the soundness of Euclid’s fifth postulate. We need not go into details here. Suffice it to say that with great skill, Saccheri worked out proposition after proposition of “Saccherian geometry” and eventually became tired of it. At one point, he decided he had reached a proposition which was “repugnant to the nature of the straight line”. That was what he had been hoping for—to his mind, it was the long-sought contradiction. At that point, he published his work under the title *Euclid Freed of Every Flaw*, and then expired.

But in so doing, he robbed himself of much posthumous glory, since he had unwittingly discovered what came later to be known as “hyperbolic geometry”. Fifty years after Saccheri, J. H. Lambert repeated the “near miss”, this time coming even closer, if possible. Finally, forty years after Lambert, and ninety years after Saccheri, *non-Euclidean geometry* was recognized for what it was—an authentic new brand of geometry, a bifurcation in the hitherto single stream of mathematics. In 1823, non-Euclidean geometry was discovered simultaneously, in one of those inexplicable coincidences, by a Hungarian mathematician, János (or Johann) Bolyai, aged twenty-one, and a Russian mathematician, Nikolay Lobachevskiy, aged thirty. And, ironically, in that same year, the great French mathematician Adrien-Marie Legendre came up with what he was sure was a proof Euclid’s fifth postulate, very much along the lines of Saccheri.

Incidentally, Bolyai’s father, Farkas (or Wolfgang) Bolyai, a close friend of the great Gauss, invested much effort in trying to prove Euclid’s fifth postulate. In a letter to his son János, he tried to dissuade him from thinking about such matters:

You must not attempt this approach to parallels. I know this way to its very end. I have traversed this bottomless night, which extinguished all light and joy of my life. I entreat you, leave the science of parallels alone. ... I thought I would sacrifice myself for the sake of the truth. I was ready to become a martyr who would remove the flaw from geometry and return it purified to mankind.

I accomplished monstrous, enormous labors; my creations are far better than those of others and yet I have not achieved complete satisfaction. For here it is true that *si paullum a summo discessit, vergit ad imum*. I turned back when I saw that no man can reach the bottom of this night. I turned back unconsolated, pitying myself and all mankind. ... I have traveled past all reefs of this infernal Dead Sea and have always come back with broken mast and torn sail. The ruin of my disposition and my fall date back to this time. I thoughtlessly risked my life and happiness—*aut Caesar aut nihil*.¹

But later, when convinced his son really “had something”, he urged him to publish it, anticipating correctly the simultaneity which is so frequent in scientific discovery:

When the time is ripe for certain things, these things appear in different places in the manner of violets coming to light in early spring.²

How true this was in the case of non-Euclidean geometry! In Germany, Gauss himself and a few others had more or less independently hit upon non-Euclidean ideas. These included a lawyer, F. K. Schweikart, who in 1818 sent a page describing a new “astral” geometry to Gauss; Schweikart’s nephew, F. A. Taurinus, who did non-Euclidean trigonometry; and F. L. Wachter, a student of Gauss, who died in 1817, aged twenty-five, having found several deep results in non-Euclidean geometry.

The clue to non-Euclidean geometry was “thinking straight” about the propositions which emerge in geometries like Saccheri’s and Lambert’s. The Saccherian propositions are only “repugnant to the nature of the straight line” if you cannot free yourself of preconceived notions of what “straight line” must mean. If, however, you can divest yourself of those preconceived images, and merely let a “straight line” be something which satisfies the new propositions, then you have achieved a radically new viewpoint.

Undefined Terms

This should begin to sound familiar. In particular, it harks back to the pq-system, and its variant, in which the symbols acquired passive meanings by virtue of their roles in theorems. The symbol q is especially interesting, since its “meaning” changed when a new axiom schema was added. In that very same way, one can *let the meanings of “point”, “line”, and so on be determined by the set of theorems (or propositions) in which they occur*. This was the great realization of the discoverers of non-Euclidean geometry. They found different sorts of non-Euclidean geometries by denying Euclid’s fifth postulate in different ways and following out the consequences. Strictly speaking, they (and Saccheri) did not deny the fifth postulate directly, but rather, they denied an equivalent postulate, called the *parallel postulate*, which runs as follows:

Given any straight line, and a point not on it, there exists one, and only one, straight line which passes through that point and never intersects the first line, no matter how far they are extended.

The second straight line is then said to be parallel to the first. If you assert that no such line exists, then you reach *elliptical geometry*; if you assert that at least two such lines exist, you reach *hyperbolic geometry*. Incidentally, the reason that such variations are still called

¹Herbert Meschkowski, *Non-Euclidean Geometry*, pp. 31–2.

²Ibid., p. 33.

“geometries” is that the core element—absolute, or four-postulate, geometry—is embedded in them. It is the presence of this minimal core which makes it sensible to think of them as describing properties of some sort of geometrical space, even if the space is not as intuitive as ordinary space.

Actually, elliptical geometry is easily visualized. All “points”, “lines”, and so forth are to be parts of the surface of an ordinary sphere. Let us write “POINT” when the technical term is meant, and “point” when the everyday sense is desired. Then, we can say that a POINT consists of a pair of diametrically opposed points of the sphere’s surface. A LINE is a great circle on the sphere (a circle which, like the equator, has its center at the center of the sphere). Under these interpretations, the propositions of elliptical geometry, though they contain words like “POINT” and “LINE”, speak of the goings-on on a sphere, not a plane. Notice that two LINES always intersect in exactly two antipodal points of the sphere’s surface—that is, in exactly one single POINT! And just as two LINES determine a POINT, so two POINTS determine a LINE.

By treating words such as “POINT” and “LINE” as if they had only the meaning instilled in them by the propositions in which they occur, we take a step towards complete formalization of geometry. This semiformal version still uses a lot of words in English with their usual meanings (words such as “the”, “if”, “and”, “join”, “have”), although the everyday meaning has been drained out of special words like “POINT” and “LINE”, which are consequently called undefined terms. Undefined terms, like the p and q of the pq -system, do get defined in a sense: implicitly—by the totality of all propositions in which they occur, rather than explicitly, in a definition.

One could maintain that a full definition of the undefined terms resides in the postulates alone, since the propositions which follow from them are implicit in the postulates already. This view would say that the postulates are implicit definitions of all the undefined terms, all of the undefined terms being defined in terms of the others.

Appendix E

Take-home Midterm Exam

In completing this exam, you may consult your notes, the textbook, and handouts from this class, but you may not consult any other sources or discuss the problems with anyone other than me.

Your final copy will not be accepted if it does not include the following signed honor pledge:

“I pledge on my honor that this examination represents my own work in accordance with University and class rules; I have not consulted any outside sources, nor have I given or received help in completing this exam.”

1. In this problem, $d_E(P_1, P_2)$ denotes the Euclidean distance between the points P_1 and P_2 , while $d_T(P_1, P_2)$ denotes the taxicab distance between them.

Let $A = (-2, 0)$, $B = (2, 0)$, and $C = (3, 1)$.

- (a) Find $d_E(A, C)$ and $d_T(A, C)$.
- (b) Either on graph paper, or else using Geometer's Sketchpad, graph the following sets, and explain how you found them:
 - i. $\{P | d_E(A, P) + d_E(B, P) = 6\}$
 - ii. $\{P | d_T(A, P) + d_T(B, P) = 6\}$
 - iii. $\{P | d_T(A, P) + d_T(C, P) = 6\}$
 - iv. $\{P | d_T(A, P) - d_T(C, P) = 2\}$

2. For each of the following statements, explain whether or not it is true on each of the sphere, the hyperbolic plane, and the Euclidean plane, and why.

- (a) A quadrilateral with at least 3 right angles has all right angles.
- (b) Given a line L and a point P not on that line, there is exactly one line through P that is parallel (in the sense of non-intersecting) to L .
- (c) Given a line L and a point P not on that line, there is exactly one line through P that is perpendicular to L .

3. Write a paragraph on each of the following. Your thoughtful answers to these questions will help to improve this course for the rest of the semester and for future semesters.

- (a) Here are some things that I think are going well in the course so far—let's keep doing them:

- (b) Here are some suggestions for the rest of the semester and future semesters:
 - (c) Here are some personal reflections on my work in this class so far:
4. We've noticed in class two different ways in which the Euclidean plane is special when compared to the hyperbolic plane and the sphere. Both of the following facts are true on the plane, but not on the other surfaces:
- A. If L and M are parallel transports along some transversal, then they are parallel transports along any transversal.
 - B. The sum of the angles in any triangle is 180 degrees.

Show that this is no coincidence by showing that, on any surface, A is true if and only if B is true.

(This is part of Problem 10.2, so you may want to look at that problem in the textbook.)

Teaching Notes:

I generally give students a week to work on this take-home midterm on their own. They turn it in and I give it a numerical grade. After this, we talk about it in class, and then I give them the opportunity to revise it to get up to half of the points they lost back. This is probably the hardest assignment of the semester.

The last problem is the first time in these courses that the students have been asked to show that two assumptions are equivalent. We generally spend some time in class after they have had a chance to work on the exam but before it is due talking about how one goes about showing such a thing. I try to help them understand in each case what they need to start with, and where they need to end up. I tell them that they can make any assumptions they want to in their proofs as long as what they are assuming is true on all three of the primary spaces we have been working with (the plane, the sphere, and the hyperbolic plane).

Teaching Notes:

This is a summary of the whole class that I sometimes give out at our last meeting to try to tie everything together.

Appendix F

Sample Class Schedule

The following is a sample schedule for a typical fifteen week semester, with classes meeting twice a week for 75 minutes each.

#	In- class activities	Assignments due
1	Class introductions; discussion of the philosophy of this class; squares on a geoboard activity	
2	Taxicab Geometry Project	Reading: How to Use this Book, pp. xxv-xxvi, and Ch. 0
3	Taxicab Geometry Project	
4	Taxicab Geometry Project	
5	Taxicab Geometry Project	Making a hyperbolic plane
6	Straightness & Postulates on Hyperbolic Plane	
7	Straightness & Postulates on Hyperbolic Plane	Taxicab Geometry (First Draft)
8	Non-Euclid	
9	Non-Euclid	Straightness & Postulates on Hyperbolic Plane
10	Finish non-Euclid; discuss folding a parallel line through a given point; start work on Problem 8.2	Folding a parallel line through a given point
11	Discuss symmetry proof of Isosceles Triangle Theorem; continue work on Problem 8.2	Taxicab Geometry (Final Draft)
12	Discuss non-Euclid; finish work on 8.2	Non-Euclid observations
13	Problem 7.2	Paper Folding/Problem 8.2
14	Problems 8.3 and 8.4	
15	Problems 8.3 and 8.4	
16	Problems 8.3 and 8.4	

#	In- class activities	Assignments due
17	Polydron activity	
18	Midterm given out; Computer Lab 1 (Paper Folding)	Problems 8.3 and 8.4
19	Computer Lab 1 (Paper Folding)	
20	Finish Computer Lab 1; start Computer Lab 2 (Products of Reflections)	Midterm
21	Pick final projects; Computer Lab 2 (Products of Reflections)	
22	Discuss Midterm; Computer Lab 2 (Products of Reflections)	
23	Start Final Projects	Computer Lab 1 (Paper Folding)
24	Parallel Postulates assignment	
25	Parallel Postulates assignment	Computer Lab 2 (Products of Reflections)
26	Final Projects	
27	Final Projects	Parallel Postulates assignment
28	KaleidoTile	Midterm revisions
29	Final Projects	
30	Final Projects	
31	Final class meeting during finals week; students present final projects	Final projects and completed portfolios

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